

# Input Form (IF)

Score: 4/5 — Minor gaps | Confidence: high

Benchmark: LAILA | Context: Arab-Country G10–12 Arabic Teachers: AI Essay Grading Assistance

**Input Form definition:** Whether the input signal encoding (text, audio, image parameters) matches deployment conditions.

Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

## Justification

The benchmark's input form is text-only Modern Standard Arabic submitted digitally via Microsoft Forms with a 65-minute time limit [Q26], producing essays averaging 171 words with most between 90 and 210 words [Q39, Q134, Q135]. PII was systematically removed and pre-generated anonymous IDs assigned [Q97]. This matches the deployment scenario directly: the elicitation marks IF as LOWER priority because deployment is text-only MSA. No script, modality, or capture-device mismatch exists. The main residual concerns are: (1) the short essay length (171-word average) is acknowledged as limiting assessment of long-form writing [Q92], and many national high-school assessments may produce longer compositions; (2) the digital-submission collection mode may not match handwritten or paper-based classroom workflows still common in some target countries — Saudi Arabia has Madrasati [WEB-17] and UAE has high digital adoption [WEB-1], but Egypt, Jordan, Lebanon, Morocco have no comparable national K-12 LMS documented in searches. The form is broadly compatible but biased toward shorter essays and digital-native workflows.

| ID     | Evidence                                                                                                                                                                                                                                                                                                           |
|--------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Q26    | "taining one persuasive and one explanatory prompt, along with a unique, pre-generated ID to maintain students' anonymity while enabling metadata tracking [D6]. We also configured a Microsoft Form for digital essay submission, set with a 65-minute time limit to enforce a time-restricted setup [D5]." (p.5) |
| Q39    | "Table 3 shows that the number of essays per prompt ranges from 500 to 1,181. Essay lengths demonstrate consistency across prompts, with an average of 171 words and a maximum of 706 words." (p.6)                                                                                                                |
| Q92    | "Fifth, with an average essay length of only 171 words, the dataset lacks extended writing samples, restricting the assessment of models on long-form academic tasks requiring more complex, extended compositions." (p.10)                                                                                        |
| Q97    | "To protect participant privacy, all submissions were assigned pre-generated anonymous identifiers, and personally identifiable information was systematically removed during data cleaning." (p.10)                                                                                                               |
| Q134   | "The data exhibits a unimodal pattern centered around an average essay length of 171 words, with the majority of essays concentrated between 90 and 210 words." (p.15)                                                                                                                                             |
| Q135   | "A pronounced peak occurs in the 150 to 180 word range, where the count reaches its highest at 1,065 essays." (p.15)                                                                                                                                                                                               |
| WEB-1  | UAE teachers ~75% AI tool adoption per OECD TALIS 2024 ( <a href="https://www.agbi.com/education/2025/10/uae-teachers-lead-world-in-classroom-ai-adoption/">https://www.agbi.com/education/2025/10/uae-teachers-lead-world-in-classroom-ai-adoption/</a> )                                                         |
| WEB-17 | Saudi Arabia's Madrasati LMS deployed to all public K-12 schools, 92% student attendance ( <a href="https://www.rowad-alkhaleej.edu.sa/en/online-learning-in-saudi-arabia/">https://www.rowad-alkhaleej.edu.sa/en/online-learning-in-saudi-arabia/</a> )                                                           |

## Strengths

- Text-only MSA matches deployment modality and Arabic script requirements [Q26]
- Authentic timed writing conditions emulate high-stakes testing [Q25]
- Systematic anonymization and PII removal supports privacy compliance [Q97]
- Comparison table situates LAILA against other AES datasets on length and annotation depth [Q12]

| ID  | Evidence                                                                                                                                                                                                                                                                        |
|-----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Q12 | "Table 2: Comparison of LAILA with existing essay datasets. "Len" is average essay length in words; "HOL" indicates holistic scoring; "Eur" covers German, Italian, and Czech; "L1/L2" denotes native or second-language learners; "Public" here means freely available." (p.2) |

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|------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Q25</b> | “[D5] Ensuring Authentic Writing Conditions: To emulate real academic or high-stakes testing scenarios, LAILA was designed to collect essays written under timed conditions using a digital submission platform. This setup ensures consis” (p.3)                                                                  |
| <b>Q26</b> | “taining one persuasive and one explanatory prompt, along with a unique, pre-generated ID to maintain students’ anonymity while enabling metadata tracking [D6]. We also configured a Microsoft Form for digital essay submission, set with a 65-minute time limit to enforce a time-restricted setup [D5].” (p.5) |
| <b>Q97</b> | “To protect participant privacy, all submissions were assigned pre-generated anonymous identifiers, and personally identifiable information was systematically removed during data cleaning.” (p.10)                                                                                                               |

## Checklist

### Checklist item definitions:

| ID          | Assessment Question                                          |
|-------------|--------------------------------------------------------------|
| <b>IF-1</b> | Compare signal distributions between source and target       |
| <b>IF-2</b> | Check if regional infrastructure supports data capture specs |
| <b>IF-3</b> | Identify domain-specific form differences                    |
| <b>IF-4</b> | Document form mismatches harming external validity           |

### Checklist responses:

**IF-1:** No script or signal-distribution mismatch — both source and target are MSA text in Arabic script [Q26]. Length distribution skews short (171-word mean, peak 150–180 [Q134, Q135]); deployment essays may be longer in some curricula, which is a partial distribution mismatch.

**IF-2:** Digital-submission infrastructure exists in Gulf states (Madrasati in Saudi Arabia, high UAE adoption) [WEB-17, WEB-1] but is less documented for Egypt, Jordan, Lebanon, Morocco; teachers in those countries may still receive handwritten essays requiring transcription before AI scoring.

**IF-3:** The CAST rubric is provided in Arabic for annotation [Q114] with English translation [Q116]; AraBERT encodes essay+prompt jointly [Q157] and AraT5 receives instruction+essay [Q148]; LLM context limit of 4,096 tokens with iterative example removal [Q184, Q185] is a relevant deployment-time form constraint.

**IF-4:** Documented form considerations: (1) average essay length is short (171 words) and may not represent longer compositions found in some curricula [Q92]; (2) no audio/handwriting modality despite handwritten classroom workflows being plausible in non-Gulf countries; (3) the digital Microsoft Forms collection [Q26] may not match all deployment workflows.

| ID          | Evidence                                                                                                                                                                                                                                                                                                           |
|-------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Q26</b>  | “taining one persuasive and one explanatory prompt, along with a unique, pre-generated ID to maintain students’ anonymity while enabling metadata tracking [D6]. We also configured a Microsoft Form for digital essay submission, set with a 65-minute time limit to enforce a time-restricted setup [D5].” (p.5) |
| <b>Q92</b>  | “Fifth, with an average essay length of only 171 words, the dataset lacks extended writing samples, restricting the assessment of models on long-form academic tasks requiring more complex, extended compositions.” (p.10)                                                                                        |
| <b>Q114</b> | “For annotating LAILA, we adopted the rubric from the Core Academic Skills Test (CAST), which is provided in Arabic, developed by the Qatar University Testing Center (QUTC).” (p.12)                                                                                                                              |
| <b>Q116</b> | “An English-translated version of the CAST rubric is presented in Table 6.” (p.12)                                                                                                                                                                                                                                 |
| <b>Q134</b> | “The data exhibits a unimodal pattern centered around an average essay length of 171 words, with the majority of essays concentrated between 90 and 210 words.” (p.15)                                                                                                                                             |
| <b>Q135</b> | “A pronounced peak occurs in the 150 to 180 word range, where the count reaches its highest at 1,065 essays.” (p.15)                                                                                                                                                                                               |
| <b>Q148</b> | “The input to the model consisted of an instruction to score the essay along with the essay text, and the model was fine-tuned to generate the trait names followed by their corresponding scores.” (p.16)                                                                                                         |
| <b>Q157</b> | “The essay and prompt were provided as input to the encoder, with the max-pooled token representation concatenated with the handcrafted features.” (p.16)                                                                                                                                                          |
| <b>Q184</b> | “The number of examples was fixed to 5, balancing scoring context with the context length limit (4096 tokens).” (p.18)                                                                                                                                                                                             |
| <b>Q185</b> | “When a prompt exceeded this limit, we iteratively removed examples until it fit.” (p.18)                                                                                                                                                                                                                          |

|               |                                                                                                                                                                                                                                                               |
|---------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>WEB-1</b>  | UAE teachers ~75% AI tool adoption per OECD TALIS 2024<br>( <a href="https://www.agbi.com/education/2025/10/uae-teachers-lead-world-in-classroom-ai-adoption/">https://www.agbi.com/education/2025/10/uae-teachers-lead-world-in-classroom-ai-adoption/</a> ) |
| <b>WEB-17</b> | Saudi Arabia's Madrasati LMS deployed to all public K-12 schools, 92% student attendance<br>( <a href="https://www.rowad-alkhaleej.edu.sa/en/online-learning-in-saudi-arabia/">https://www.rowad-alkhaleej.edu.sa/en/online-learning-in-saudi-arabia/</a> )   |

### Information Gaps

- Whether handwritten-essay workflows remain dominant in non-Gulf target countries
- Length distribution of typical G10–12 essays in Egyptian, Jordanian, Lebanese, Moroccan curricula

### Requires Expert Verification

- Country-by-country prevalence of digital vs. paper essay submission in G10–12 Arabic classrooms